

Air Pollution in Lombardy, Italy

Bayesian Data Analysis

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The region of Lombardy, situated in northern Italy between the Alps and tributaries of the river Po, is known to be one of the most polluted areas of Europe [1]. In this report we concern ourselves with air pollution made up of fine particulate matter less than $2.5\mu\text{m}$ in diameter ($\text{PM}_{2.5}$), an incredibly hazardous form of pollution often associated with premature mortality. In particular, we will investigate and attempt to elicit factors that drive $\text{PM}_{2.5}$ levels in Lombardy, including meteorological effects, livestock, pollutant emissions, and local vegetation.

Our report is structured as follows. Section 1 describes the data we use and explains our choice of covariates. Then, Section 2 outlines the three statistical models we will use, with these models being compared and their performance analysed in Section 3. Finally, we will provide an interpretation of these models in Section 4, discussing the statistical significance of factors associated with $\text{PM}_{2.5}$.

1 Data: Agrimonia

Agrimonia (AgrImOnIA; Agriculture Impact On Italian Air) is an open access dataset containing information about air quality, weather, livestock, emissions, and land use between 2016 and 2021. Importantly, the data is grouped by 141 air quality stations within Lombardy when a 0.3° (approx 30km) buffer is applied to the border. More information on the building of this dataset is described in [2].

To reduce the size of the data, and to avoid inconsistent trends arising from the COVID-19 pandemic influencing our analysis, we only consider data from the years 2016-2019 inclusive.

1.1. Exploratory data analysis. First, we explore the response variable $\text{PM}_{2.5}$. It's measured by concentration, $\mu\text{g}/\text{m}^3$, and unfortunately not all air quality stations reliably report it. For robust analysis, we only consider stations with at least 50% uptime of $\text{PM}_{2.5}$ reporting, leaving us with 50 stations within the greater Lombardy area.

Of these stations, there is a noticeable temporal trend in $\text{PM}_{2.5}$ levels. Indeed, it typically rises and spikes in the colder, winter months, and falls in the warmer, summer months. Furthermore, there seems to be strong spatial dependence on $\text{PM}_{2.5}$ levels, with stations reporting high quantities often clustering closely together. See Figure 1. Levels of $\text{PM}_{2.5}$ fall between $0\mu\text{g}/\text{m}^3$ and $182\mu\text{g}/\text{m}^3$, generally being fairly right-skewed. An important modelling assumption we make in this analysis is that, under a log transform, the levels of $\text{PM}_{2.5}$ are normally distributed.

Other data is far more complete, with only a small number of missing values for swine and bovine data, and no apparent missing data for meteorological, land use, or emissions data.

1.2. Choice of covariates. We aim to use a broad range of covariates from the Agrimonia dataset for better exploration of factors that drive $\text{PM}_{2.5}$. It is a sensible assumption that weather effects such as temperature, wind speed, humidity, and precipitation impact air pollution, so these are included. When the same effect is measured in more than one way, such as the mean, mode, and max wind speed at 10m and 100m above the surface of the Earth, one representative is taken to avoid high collinearity between covariates. The maximum boundary layer height is also included

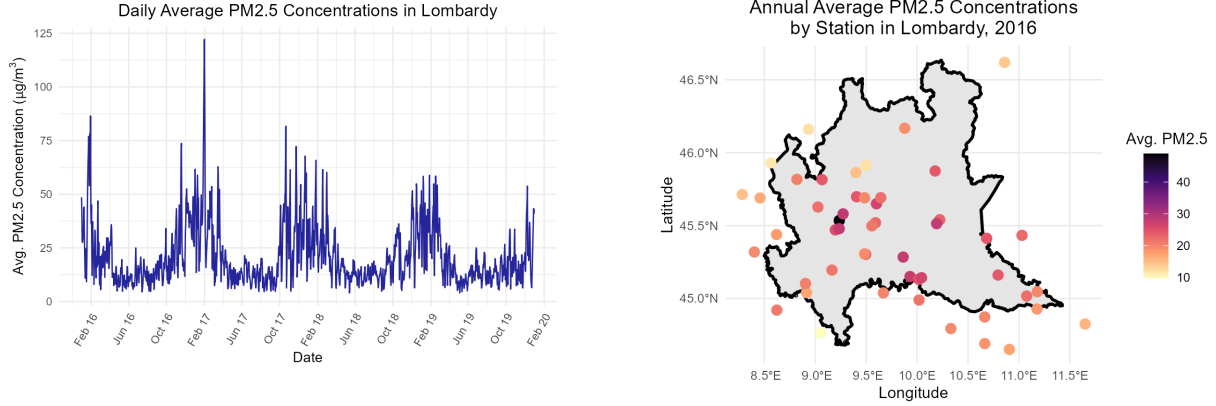


Figure 1: (Left) Weekly average $\text{PM}_{2.5}$ concentrations $\mu\text{g}/\text{m}^3$ in Lombardy between January 2016 and December 2019. (Right) Annual average $\text{PM}_{2.5}$ concentrations in Lombardy, 2016, where each dot is a monitoring station.

after demonstrating strong correlation with $\text{PM}_{2.5}$ in the exploratory data analysis, as is the altitude of the monitoring station.

With regards to livestock, we consider the density of both swine and bovines in the region. Agriculture and livestock are known to be large sources of ammonia release into the atmosphere, but we also supplement our model with the total sum of ammonia emissions to account for human activity or other factors. Finally, we also consider the impact of high and low vegetation, such as trees and bushes respectively, as it is largely agreed that these have an impact on air pollution (e.g. [3]). A summary of these covariates is given in Table 1.

Variable	Description	Units
temp_2m	Temperature at 2m above surface.	$^{\circ}\text{C}$
ws_10m_mean	Mean wind speed at 10m above surface.	m/s
rh_mean	Mean of relative humidity.	%
tot_precip	Total precipitation, daily.	m
blh_max	Maximum boundary layer height.	m
altitude	Altitude of monitoring station.	m
ammonia_sum	Sum of all ammonia (NH_3) emissions.	mg/m^2
swine_dens	Density of swines by monitoring station.	$\#/\text{km}^2$
bovine_dens	Density of bovines by monitoring station.	$\#/\text{km}^2$
high_veg	High vegetation (e.g. trees) by station.	m^2/m^2
low_veg	Low vegetation (e.g. bushes) by station.	m^2/m^2

Table 1: Description of model covariates. Naming conventions here are similar to that of the original Agrimonia dataset, with more information accessible in its accompanying paper [2].

1.3. Preprocessing. Here we will briefly describe the preprocessing steps taken. As already discussed, we only consider the years 2016-2019 inclusive, and only stations with at least 50% uptime of $\text{PM}_{2.5}$ monitoring. Additionally, any levels of $\text{PM}_{2.5}$ less than $0.25\mu\text{g}/\text{m}^3$ are considered as erroneous and set to missing.

New variables that indicate daily, weekly, and month indexes are created, together with numeric

station identifiers. For better interpretation of our model, and to potentially aid numerical stability, all covariates described in Table 1 are scaled. Furthermore, as also discussed previously, we take the log transform of PM_{2.5} levels, and assume the result is distributed normally.

Finally, monitoring stations are split into folds for future cross-validation purposes. We employ a 5-fold (groups of 10) and 25-fold (groups of 2) splitting strategy of monitoring stations with a set seed for reproducibility. These choices are discussed further in Section 3.

2 Model Specification

Our investigation employs three Bayesian statistical models to model the concentration of PM_{2.5} levels in Lombardy, all implemented with R-INLA [4, 5]. We assume the data are realisations of a random spatiotemporal process, where discrete station measurements act as noisy observations of a continuous, underlying latent field. These measurements are also proposed to be influenced by the factors described in Table 1.

In general, let s denote an arbitrary monitoring station, with $\{s_i\}$, $i = 1, 2, \dots, 50$ representing the set of all 50 monitoring stations. Let t be a discrete time index for the daily observations of (log transformed) PM_{2.5} levels $y(s, t)$. Our general model framework is then

$$y(s, t) = R(s, t) + D(s, t) + \varepsilon(s, t), \quad (1)$$

where $R(s, t)$ is a regression model including the covariates in Table 1, $D(s, t)$ describes the spatiotemporal effects, and $\varepsilon(s, t)$ is noise accounting for measurement and modelling errors, assumed to arise from a zero-mean normal distribution.

A penalised complexity prior is placed on the precision $\tau = \sigma^{-2}$ of the errors ε , with $P(\sigma > 1) = 0.025$; that is, an error of 100% is unlikely (probability of 2.5%). This choice was employed on all of our models for a few reasons. First, it is difficult to elicit the scale of the error when working with unscaled spatiotemporal components and scaled regressors. Furthermore, in the spirit of this investigation’s objective, it is preferred that the model attempts to place weight on its predictors rather than attributing it to noise.

2.1. Model description. Here we describe the three models implemented with R-INLA. Choice of priors is discussed, and will be discussed further in Section 2.2 (prior sensitivity).

2.1.1. Model one; spatial linear regression. Our first model assumes a constant spatial field across the entire timeline, with linear fixed effects arising from the covariates discussed. Formally,

$$y(s, t) = \beta_0 + \sum_{k=1}^K \beta_k X_k + D(s) + \varepsilon(s, t), \quad (\text{M1})$$

where β_0 is the intercept term, X is the covariate matrix, $D(s)$ is the constant spatial field, and $\varepsilon(s, t)$ the noise. To account for the lack of temporal element in D , the covariates include those in Table 1 together with new monthly indicator variables.

The spatial field $D(s)$ is a recurring theme in our models, implemented because of the potential for strong spatial correlation as discussed in Section 1. It is a continuous Gaussian random field, approximated by INLA as a Gaussian Markov random field using the SPDE approach [4]. We employ the same mesh across all three of our models, and due to computational limitations arising from model three, only a moderate number of mesh points ($n = 242$) are used.

Penalised complexity priors are placed on the range and variance of the spatial field as follows. For the range, we suggest it is unlikely that spatial correlation diminishes before 0.5° (approx 50km). This is reasonable, given the greater Lombardy region is approximately 300km in diameter, and in line with other studies [6]. Additionally, we suggest it is unlikely that the variance given by the spatial field is greater than the variance present in the data, $\sigma \approx 0.75$. In both cases, ‘unlikely’ is quantified as a 5% chance, providing a modest amount of uncertainty.

With respect to the regression coefficients, we place flat, uninformative priors $\beta_k \sim \mathcal{N}(0, 1000)$ for $k = 0, 1, \dots, K$. It should be noted that these are the default priors used by INLA, but perfectly sufficient for our analysis.

2.1.2. Model two; spatial nonlinear regression. This model extends the first model by incorporating nonlinear effects for select covariates. In particular, we assume the temperature, wind speed, and maximum boundary layer height assert nonlinear influence on $\text{PM}_{2.5}$, i.e.

$$y(s, t) = \beta_0 + \sum_{k=1}^{K-3} \beta_k X_k + \sum_{k=K-2}^K f_k(X_k) + D(s) + \varepsilon(s, t), \quad (\text{M2})$$

where the covariates in Table 1 are ordered such that `temp_2m`, `ws_10m_mean`, and `blh_max` correspond to indices $K-2, K-1, K$.

The same priors for the spatial field and the regression coefficients are used as in model one, and here the smoothing functions are modelled using a first order random walk (RW1) prior on $B = 50$ bins of the covariate values. We place a harsh penalised complexity prior on the precision of this random walk to prevent overfitting, with this being discussed further in Section 2.2.

2.1.3. Model three; spatiotemporal linear regression. Our third and final model also extends the first model, this time incorporating a time component to the spatial field, i.e.

$$y(s, t) = \beta_0 + \sum_{k=1}^{\tilde{K}} \beta_k X_k + D(s, t) + \varepsilon(s, t). \quad (\text{M3})$$

This is implemented in R-INLA as a first order autoregressive process with temporal correlation coefficient ρ , where time is measured weekly. With the notion of time now being captured by the spatial field, the monthly indicator variables are no longer used. Stations are now treated as i.i.d, with a harsh penalised complexity prior on the precision.

Again, we place penalised complexity priors on the autoregressive temporal component. We suggest that variance across weeks exceeds 100% with a small probability of 2.5%. The prior for the correlation coefficient ρ is assumed zero as the base model in order to match the priors on the monthly indicator variables in model one and two, but with a moderate probability that $\rho > 0$. Specifically, we use INLA’s `pc.cor0` with a 50% probability that $\rho > 0.5$.

2.2. Prior sensitivity. First, we consider the sensitivity of the spatial field range prior in models one and two. Let d_0 denote the distance at which we are 95% confident spatial correlation does not diminish, measured in degrees ($1^\circ \approx 100\text{km}$). Across moderate values $d_0 \in \{0.1, 0.3, 0.5\}$, the posterior mean for the spatial range was stable, landing between 0.32 and 0.38. Even with an extreme value of $d_0 = 1$, the posterior mean only increased to 0.44. Moreover, the standard deviation of the spatial range was almost invariant across d_0 , always equal up to one decimal place ($\sigma = 4.1$). This indicates that the spatial field is relatively robust under prior specification, and that the choice of $d_0 = 0.5$ is indeed reasonable.

Next, we consider the penalised complexity prior on the smoothing functions f_k in model two. These assert that the change in effect across each of the $n = 50$ bins is unlikely (we choose a 1% chance) to be more than U_0 , where we take $U_0 \in \{0.01, 1, 10000\}$ labelled as ‘strictest’ (highly regularised), ‘stricter’, and ‘flexible’ (almost unconstrained) respectively. Recall that we are on the log scale, so U_0 denotes a multiplicative change. Figure 2 depicts the posterior of each nonlinear effect with these priors in place. All three regressors exhibit reasonably smooth curves with a large amount of overlap, even without regularisation from the prior. Stricter choices form a smooth posterior while still exhibiting nonlinearity, and the credible intervals of the effects shrink.

Importantly, the large overlap and lack of catastrophic overfitting indicate that these curves are driven by the data and robust to the choice of prior. We choose the strictest prior because it captures the nonlinearity well while providing far desirable variance, especially for extreme values of wind speed and boundary layer height.

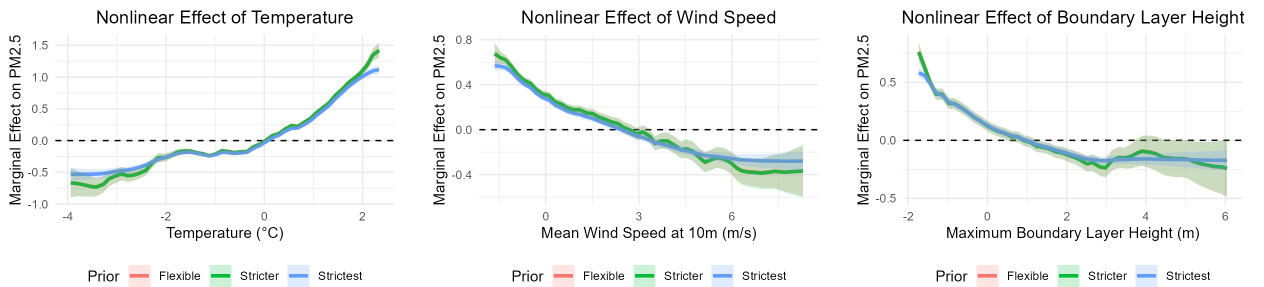


Figure 2: The posterior of nonlinear effects in model two with different penalised complexity priors on the random walk smoothing function.

3 Model Comparison and Validation

The performance of each of the models described in Section 2 is described here.

3.1. Performance metrics and residuals. First, we discuss each models performance when applied on the entire dataset, with Table 2 reporting the DIC, WAIC, and information about relevant hyperparameters. A fitted vs residuals plot for each model is also given in Figure 3.

	Model 1		Model 2		Model 3	
DIC	102047.13		99449.40		73848.52	
WAIC	102049.92		99455.83		73987.47	
Hyperparameter	Mean	95% CI	Mean	95% CI	Mean	95% CI
Residuals Precision	3.792	[3.751, 3.832]	3.942	[3.900, 3.984]	5.953	[5.888, 6.020]
Spatial Range	0.383	[0.255, 0.553]	0.357	[0.261, 0.566]	3.125	[2.951, 3.318]
Spatial SD	4.097	[3.289, 5.050]	4.113	[3.317, 5.108]	1.039	[0.926, 1.183]
Temporal Correlation	—	—	—	—	0.966	[0.958, 0.974]

Table 2: Scores reporting the performance of each model when trained on the full data set.

An immediate observation is the significant decrease in the DIC and WAIC scores as the models progress. Indeed, model two sees a moderate improvement over model one, indicating that the nonlinear impact of temperature, wind speed, and boundary layer height do better capture the

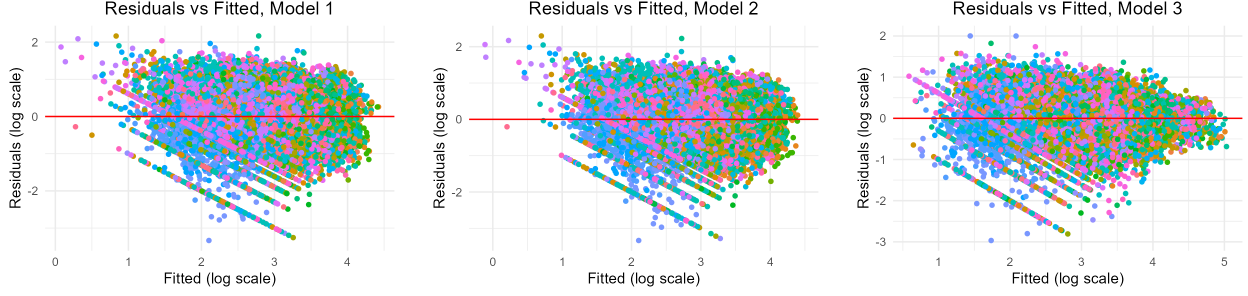


Figure 3: Residuals plot for each model, in the log scale. Points are coloured by the monitoring station ids.

dynamics at play than purely linear terms. However, most notably, model three’s DIC and WAIC scores is significantly lower than that of the first two models, despite a large number of new hyperparameters being introduced from the spatiotemporal field. This suggests that the weekly temporal trends play a large role in the model, as supported by the mean correlation coefficient $\rho = 0.966$, and possibly that the monthly indicator variables in the static spatial models are not enough to describe the temporal element of $\text{PM}_{2.5}$ levels.

Another interesting observation here is the expansion of the posterior spatial range from the first two models into the third. It’s possible that the absence of a strong temporal parameter is forcing the spatial field to shrink because time-related dynamics are not being accounted for; or, on the contrary, it’s possible that the high temporal persistence in the third model dominates the influence of the spatial field and allows its range to grow very large.

Residuals of all three models are reasonably scattered about zero, becoming slightly tighter with each model. The observed diagonal lines is likely an artifact of the data; air quality monitoring stations do not provide continuous measurements, but integer values for the concentration $\mu\text{g}/\text{m}^3$. Thus, even under a log transformation, our data takes a small collection of values, meaning diagonal lines on the residual plots are acceptable.

Residuals of each station cluster together in some instances, but generally mix quite well, with no one station causing a consistent pattern of outliers.

3.2. Cross validation. To further assess each model, we measure their predictive performances with a variant of leave-one-out cross validation where entire stations’ worth of data are withheld in training. In particular, we split the 50 stations into (a) groups of 10, for 5-fold cross validation, and (b) groups of 2, for 25-fold cross validation. Groups are determined with a seed for reproducibility, and each model is assessed on the same data subsets.

By withholding a large portion of training data, it’s possible that the spatial structure of the model is disrupted, and thus it is expected that the models perform better on the 25-fold validation than the 5-fold. Ideally, one station is withheld from training, but we are limited by the computational demand arising from the spatiotemporal model (model three).

After the predictions are made, we compute the RMSE, MAE, and R^2 values for each model. These are given in Table 3, and it is noted that these metrics are on the same scale as the original data.

The general trend here is the same as given by the performance metrics previously discussed. Model three far outperforms the first two, with a moderately competitive R^2 score of 0.6110 when two entire stations are withheld from training. For a reasonably simple model (only linear regression in

Metric	5-Fold CV			25-Fold CV		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
RMSE	15.36	16.15	10.78	113.39	125.59	10.38
MAE	10.50	10.90	7.17	22.17	23.27	6.93
R^2	0.2530	0.2340	0.5855	0.0039	0.0026	0.6110

Table 3: Cross-validation performance metrics for each model (2dp; R^2 to 4dp).

the covariates, a weekly temporal element, and a reasonably coarse mesh), it captures the dynamics of $PM_{2.5}$ concentrations in Lombardy to a respectable degree. On the other hand, both of the first two models see a catastrophic fall in their predictive performance with 25-fold cross validation. This is unexpected, and may indicate that these models are greatly overfitting on the available data - and thus have improved performance when less data is available in the 10-fold validation.

4 Interpretation of Results

Finally, here we interpret the results of each model and look to answer our main research objective: what are the factors that drive $PM_{2.5}$ levels in Lombardy?

To begin, we report the 95% credible intervals for each of the covariates described in Table 4. The nonlinear effects of model one are reported in Figure 4; these are the same as the curves with strictest penalised complexity discussed earlier, but reported again for completeness.

Variable	Model 1		Model 2		Model 3	
	95% CI	Significant	95% CI	Significant	95% CI	Significant
temp_2m	[0.32, 0.34]	Yes	–	–	[0.50, 0.54]	Yes
ws_10m_mean	[-0.16, -0.15]	Yes	–	–	[-0.13, -0.12]	Yes
rh_mean	[0.05, 0.07]	Yes	[0.10, 0.11]	Yes	[0.13, 0.14]	Yes
tot_precip	[-0.13, -0.12]	Yes	[-0.14, -0.13]	Yes	[-0.10, -0.09]	Yes
blh_max	[-0.18, -0.17]	Yes	–	–	[-0.13, -0.12]	Yes
altitude	[-0.45, -0.24]	Yes	[-0.44, -0.24]	Yes	[-0.14, -0.01]	Weakly
ammonia_sum	[-0.07, -0.05]	Yes	[-0.07, -0.06]	Yes	[-0.05, -0.01]	Weakly
swine_dens	[0.01, 0.11]	Weakly	[0.00, 0.11]	No	[0.06, 0.21]	Yes
bovine_dens	[-0.33, -0.18]	Yes	[-0.32, -0.17]	Yes	[-0.15, 0.02]	No
high_veg	[0.04, 0.06]	Yes	[0.11, 0.14]	Yes	[-0.04, 0.05]	No
low_veg	[-0.23, -0.20]	Yes	[-0.20, -0.18]	Yes	[-0.28, -0.20]	Yes

Table 4: The 95% credible interval (2dp), and a notion of significance, for each of the chosen covariates. Values are on the log scale.

Both model one and model two used monthly indicator variables, with each month for both models being deemed significant. In fact, an incredible amount of weight is put on these variables, with coefficients being up to ten times greater than the other covariates. This suggests the primary driver of $PM_{2.5}$ levels in the region is found in seasonal shifts, and does agree with the high level of temporal correlation found in Table 2. So much weight being placed on the monthly indicator variables also supports the notion that these models are overfitting, and helps explain why their

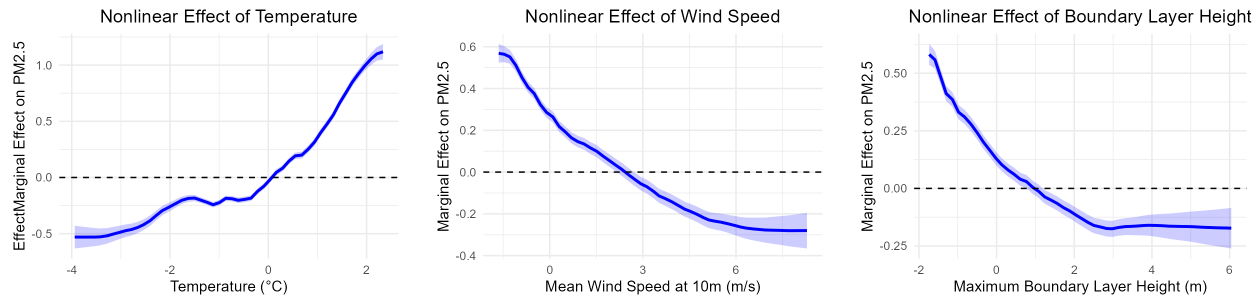


Figure 4: The posterior effects of the nonlinear covariates in model two.

predictive performance of the daily $\text{PM}_{2.5}$ levels was so poor.

With regards to the covariates, we will focus mainly on the results given by model three, the most reliable model. All weather effects including temperature, wind speed, humidity, and precipitation are suggested to be statistically significant drivers of $\text{PM}_{2.5}$. Likewise, so is the maximum boundary layer height, swine density, and low vegetation in the area - but, surprisingly, the model does not suggest that the related bovine density nor high vegetation is statistically significant. There is reasonable doubt on the influence of station altitude and the total amount of ammonia emissions, although the latter could be explained by its potential correlation with the swine/bovine density or other agricultural features.

5 Conclusions and Future Work

In conclusion, we have introduced and compare three Bayesian statistical models to investigate factors driving air quality (via $\text{PM}_{2.5}$ levels) in Lombardy, Italy. Our analysis suggests that a multitude of weather effects, boundary layer height, swine density, and low vegetation levels are statistically significant factors driving $\text{PM}_{2.5}$, and that a large portion of the variance can be explained by seasonal trends.

Our spatiotemporal model with linear regression proved to be most competitive, but there is certainly room to extend it. For example, there is evidence of nonlinearity with certain covariates (e.g. Figure 4) that could be incorporated. Moreover, a finer mesh could be used to aid spatial influence, or more informative priors can be enforced - perhaps elicited from other studies of the region.

References

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